

Transistor-Level Design and Validation of a Silicon-Carbide JFET Intel-4004-Compatible 4-bit Processor

Christian Garry* and Alton Horsfall†

Abstract—Silicon Carbide (SiC) Junction Field Effect Transistors (JFETs) are a leading candidate for digital logic operating in radiation environments and at temperatures beyond the reach of bulk silicon. To the authors’ knowledge, this work presents the first transistor-level SiC JFET implementation of an Intel-4004-compatible 4-bit microprocessor architecture, comprising 5546 transistors across the ALU, instruction decoder, program counter, three-level stack, and sixteen-register scratch pad, with a compatible 4001-style ROM model used for program execution. The contributions are three. First, a complete 4004-compatible transistor-level SiC JFET RTL design captured at the schematic level in LTspice. Second, a program-level co-validation methodology in which a custom Python reference emulator matches the LTspice CPU register-for-register at every instruction boundary, demonstrated on a bitwise AND subroutine from the MCS-4 programming manual and a 16-bit floating-point multiplication. Third, measured NAND/NOR primitive switching behaviour consistent in timing and waveform shape with LTspice simulation, characterised from 1 Hz up to 800 kHz (NOR) and 1 MHz (NAND), with an unresolved 1.5–2 V baseline offset and other limitations of the present validation chain treated explicitly.

Index Terms—Silicon Carbide (SiC), Junction Field Effect Transistors (JFETs), Radiation-Environment Electronics, High-Temperature Electronics, Intel 4004 Architecture, Resistor-Transistor Logic, Circuit Simulation, Emulator Co-validation

I. INTRODUCTION

COMPUTING in radiation- and temperature-extreme environments — fusion reactors, geothermal boreholes, deep-space probes — is constrained by the failure of silicon-based logic above 200–400 °C and under heavy ionising radiation. Wide-bandgap semiconductors offer a path forward, with Silicon Carbide (SiC) the most mature of the candidate materials: 4H-SiC has a bandgap of 3.26 eV [1], compared to 1.12 eV for silicon [2], and a high atomic-displacement threshold energy (24 eV for carbon and 66 eV for silicon in 4H-SiC [3], against 21 eV in bulk Si [4]). SiC Schottky-barrier diodes have been measured under proton [5], electron [6] and light-ion [7] irradiation without deviation from linear charge-collection behaviour, making the material suitable for both fusion and off-world applications.

Among SiC active devices, the junction field-effect transistor (JFET) is uniquely suited to high-temperature digital logic: it relies on a buried p-n junction rather than a

metal-semiconductor interface, avoiding the Schottky-barrier degradation that disqualifies MESFETs above $\sim 300^\circ\text{C}$ [8]. SiC JFETs have been demonstrated as discrete devices at $\sim 1000^\circ\text{C}$, with the main remaining degradation mechanism being interdiffusion of the metal ohmic contacts rather than wafer-level failure [9]. Complementary JFET (CJFET) logic gates have been built and measured to 623 K [10] and simulated to 500 °C [11]; the analysis in this paper instead uses resistor-transistor logic (RTL), since CJFET requires complementary p- and n-type devices and the available SiC JFET technology is single-polarity. RTL maps naturally to resistor-loaded pull-up/pull-down structures on single-polarity SiC JFETs. The comparison with the Intel 4004 is architectural rather than device-technological: the original 4004 was implemented in silicon-gate PMOS, whereas the present work implements a 4004-compatible architecture using SiC JFET RTL.

The Intel 4004 was chosen as the target architecture because of its small but complete ISA — 46 instructions, a 12-bit program counter, a 3-level subroutine stack, sixteen 4-bit scratch-pad registers, and an ALU restricted to 4-bit addition and accumulator rotation [12], [13]. This is large enough to be a realistic test of an integrated SiC JFET design (the present implementation totals 5546 JFETs) yet small enough that every subsystem can be hand-designed, validated against a Python reference emulator, and reasoned about in a single paper.

Table I situates this work against the existing SiC JFET digital-logic literature. To the authors’ knowledge, this is the first transistor-level SiC JFET implementation of an Intel-4004-compatible 4-bit microprocessor architecture; prior demonstrations have been at the logic-gate or small-IC scale. The contribution of this paper is therefore not a new device technology but a complete architecture-scale design captured in LTspice, paired with a validation methodology — register-level co-simulation against a Python reference emulator and measured-versus-simulated agreement of the primitive gates — that gives independent evidence that the simulated processor is functionally correct.

The paper is organised as follows. Section II describes the SiC JFET RTL logic family and the decomposition of the 4004 architecture into subsystems. Section III describes the design and validation methodology, including the LTspice transient simulation, the Python reference emulator, and the assembler/PWL toolchain. Section IV presents the program-level register-state co-validation and the measured-versus-simulated primitive-gate results. Section V delineates the limitations of

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TABLE I: SiC JFET digital-logic literature against which this work is positioned. “Validation” distinguishes simulation-only demonstrations from those backed by measured silicon. Operating temperature is the highest at which the cited work reports successful logic-gate operation; for this paper it is room temperature, since the device model used for the LTspice CPU simulations is not temperature-scaled.

Reference	Process / device	Logic family	Complexity	Max validated T	Validation
Zolper [8] (1998)	GaAs / SiC / GaN JFETs	–	review only	–	review
Raynaud [14] (1997)	6H-SiC buried-gate JFET	–	single device	300 °C	measured (device)
Neudeck [15] (2000)	6H-SiC JFET	RTL	individual gates	600 °C	measured
Neudeck [9] (2018)	4H-SiC JFET	RTL	small digital ICs	~1000 °C	measured
Habib [11] (2013)	4H-SiC CJFET	CJFET	individual gates	500 °C	simulation
Kaneko [10] (2022)	4H-SiC CJFET	CJFET	logic gate	350 °C (623 K)	measured
This work	4H-SiC JFET (model)	RTL	4-bit CPU (5546 JFETs)	27 °C (model)	LTspice + emulator + measured primitives

the work, including the room-temperature device model and the measured voltage offset, and Section VI concludes.

II. THEORY

In this paper, “4004-compatible” means ISA-level and program-level compatibility with the documented MCS-4 instruction semantics under the modelled 4001-style ROM interface. It does not imply pin-level, voltage-level, cycle-timing, fabrication-process, or electrical-bus equivalence with the original Intel 4004. In particular, the data-bus implementation described in Section II-B is adapted to the SiC JFET RTL logic family rather than copied electrically from the original PMOS device.

A. Logic Primitives

The LTspice model of the JFET was used to create a logic primitive. This was extended to form more complex Logic Gates. Applying a voltage to the Gate of a JFET causes the depletion region to increase. This narrows the conducting channel increasing the resistance, which decreases the current that can pass through it. If the source of the JFET is connected to a voltage source through a resistor there will then be a change in voltage at the JFET’s source when a voltage is applied to the Gate. This does not provide a large enough voltage swing between high and low and has implications for the fan out of the gate. Therefore, a voltage follower was used which works as a voltage level shifter and shifts the output voltage back to input levels of the logic gate (High: -0.8 V, Low: -3.6 V). These values were selected as they kept

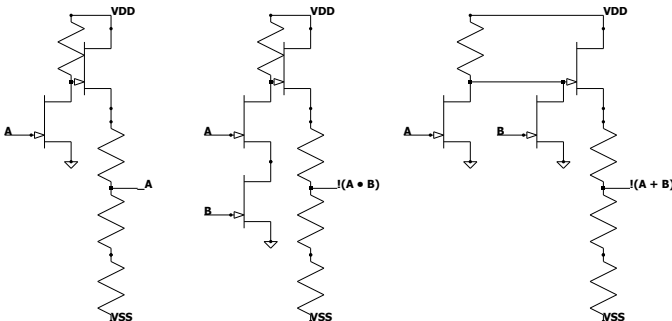


Fig. 1: Circuit diagrams of NOT, NAND, and NOR gates respectively

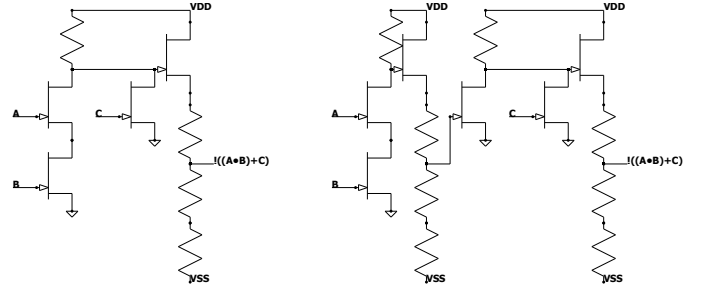


Fig. 2: Circuit diagram of a condensed boolean gate (Left) and a combination of base gates (Right), illustrating the transistor reduction for the boolean equation $(A \text{ AND } B) \text{ NOR } C$

the output voltage from decaying or oscillating when used as inputs for subsequent gates whilst using standard resistor values.

The design of more complex gates follows on logically from this (Figure 1). To design the NOR gate a second JFET can be connected to the voltage follower in parallel with the first JFET which provides two inputs both with equal function. This means that if either JFET has a voltage applied, the output of the logic gate is the same. The NAND gate was developed using similar reasoning. If both inputs are required to be on to turn the output off, connecting two JFETs in series to the voltage follower will require that both JFETs are sinking current for the output to change. Condensed boolean gates are gates that have an entire Boolean expression compressed into a single gate in order to optimise the number of transistors used (Figure 2). For example, a NOR gate in which one input JFET is replaced by two JFETs in series to ground produces $\overline{AB + C}$ while saving two devices compared to the equivalent NAND/NOR composition. This is analogous to the use of complex gates in integrated-circuit logic design, where several Boolean operations are collapsed into a single transistor network to reduce device count.

The nominal electrical parameters used throughout this paper are summarised in Table II. NAND and NOR gates were built in the lab and compared to the simulated devices, characterised from 1 Hz up to a maximum of 1 MHz; results are presented in Section IV.

TABLE II: Nominal SiC JFET RTL simulation parameters, as defined in `cpus/4004/config.py`.

Parameter	Value
Positive supply V_{POS}	+24 V
Negative supply V_{NEG}	-20 V
Logic high	-0.8 V
Logic low	-3.6 V
Threshold (digital analyser)	-2.5 V
Pull-up resistor R_1 (INV/NAND2/NOR2)	50 k Ω
Voltage-follower upper R_2	1 k Ω
Voltage-follower lower R_3	4.5 k Ω
JFET gate-source / gate-drain C_{gs0}, C_{gd0}	16.9 pF
Simulation temperature	27 °C
Target clock frequency	100 kHz

B. CPU Design

The Intel 4004 can be decomposed into several distinct sub-systems. The arithmetic logic unit (ALU), the instruction register and decoding, the stack and program counter (PC), the Scratch Pad, and the circuits related to the pins and interfacing with other external components. The specific instructions relating to each sub-system were evaluated in detail to precisely match the functionality of the original Intel 4004.

1) *ALU*: From the architecture diagram (Figure 3) it can be seen that the ALU (Figure 4) has two 4-bit registers connected to it, the Accumulator (Acc) and the Temporary (Temp) register. From the instruction set it can be seen that the ALU consists only of a 4-bit adder. The contents of the accumulator can also be shifted to the left or right by one bit, however it also includes the carry flag in this shift operation. For example, shifting right would place the least significant bit of the accumulator into the carry and what was in the carry into the most significant bit of the accumulator. Based on the instruction set it can be deduced that there is no inverter on the accumulator and that for subtraction and the other inversion instructions, for example complement accumulator (CMA), there must be an inverter between the temporary register and its connections to the bus and adder. CMA is then accomplished by transferring the value from the Accumulator to the Temp register, inverting the Temp register and transferring this inverted value back to the Accumulator. The carry flag also needs to be able to be inverted. There is a separate incrementer circuit which adds one to the temporary register. This is because the instruction Increment (INC), increments the value stored in a specific Scratch Pad register without effecting the accumulator or the carry flag of the ALU. There is not enough time in the execute portion of the instruction cycle for the accumulator value to be moved to some intermediate location to be moved back after the increment instruction is completed. This means that there has to be a separate circuit for incrementing Scratch Pad registers.

A two-stage carry flag is required because the carry and accumulator do not latch their new values simultaneously, producing a race condition between the sum write-back and the carry update. The first flip-flop captures the new carry without affecting the adder input, and the second stage is loaded from the first once the accumulator has been written. The full derivation is given in Supplementary Note S1.

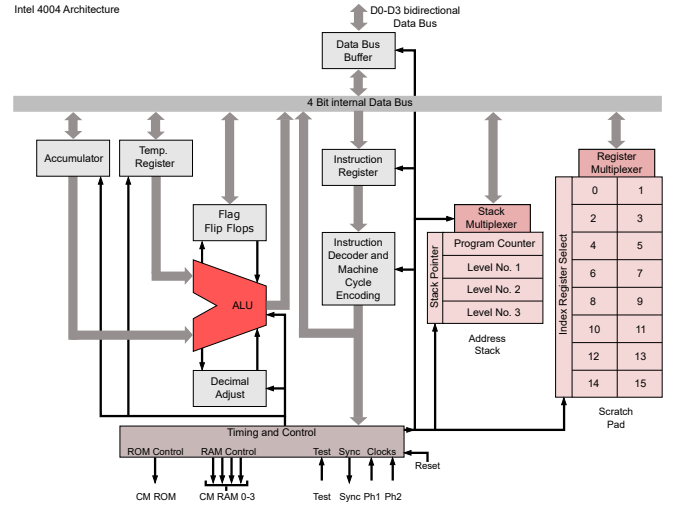


Fig. 3: Architecture of the Intel 4004 [16] showing the layout of registers and connections between CPU sub-systems

2) *Instructions*: The instruction register of the 4004 is 16-bits wide (Figure 5). Instructions are one or two eight-bit words; the most significant four bits are an opcode and the remaining bits encode an operand or, in a small number of cases, an opcode discriminator (for instance, the LSB of the lower nibble distinguishes FIN from JIN). The high eight bits are de-multiplexed to produce a one-hot instruction-selected line that, when AND-ed with the current micro-instruction step counter (Figure 7), generates each of the per-step control-line activations that constitute a micro-instruction (Figure 6). Each instruction is executed across three named phases — address ($A_{1,2,3}$), memory ($M_{1,2}$) and execute ($X_{1,2,3}$) — under which the CPU drives the address onto the four-bit external bus in the order required by the 4001 ROM and the 4002 RAM, captures eight bits of returned data into the instruction register, and

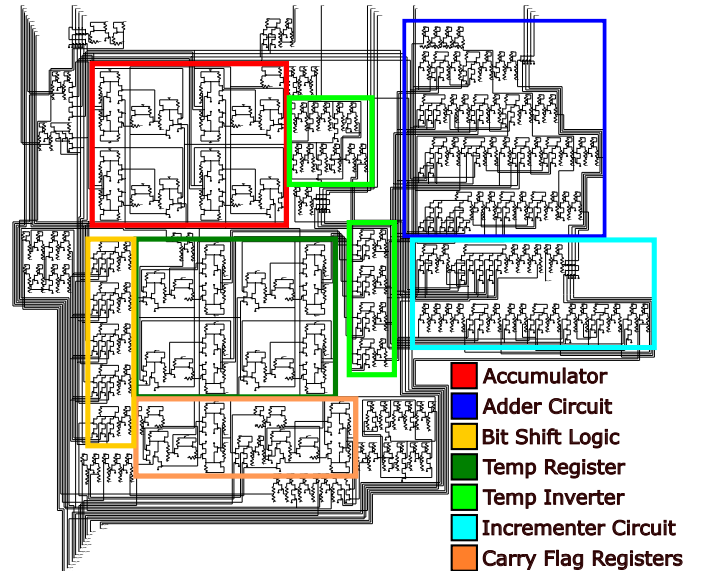


Fig. 4: Circuit diagram for the Arithmetic and Logic Unit with colour-coded sub-circuits

finally asserts the subsystem control lines required to complete the operation. Two-word instructions are handled by a single toggle flip-flop on the CPU that re-enters the lower-word load path on the next fetch cycle. The full phase-by-phase enumeration, including the ROM-side handshake, is given in Supplementary Note S2.

3) *Program Counter and Stack*: The Program Counter (PC) (Figure 8) consists of a 12-bit register and an incrementing circuit. The PC also needs to be able to write values to and from the Bus. The conditional jump (JCN) instruction indicates that each 4-bits in the PC need to be individually controlled as the high order 4-bits are not changed when JCN is performed. The limit of three micro-instruction steps for executing instructions means that the stack (Figure 9) has to have a direct connection to the PC as all 12-bits need to be written, along with additional steps related to the stack level control, within three clock pulses. This is not possible with the 4-bit bus unless the PC and Stack have a direct connection. The stack is three layers deep and needs to keep track of which level it is on. This is done using a shift register which is able to be shifted up and down to select each level. The output of each register of the stack is fed into an AND gate along with the level control line that corresponds to it so that the register only outputs when its level is selected and the "Output Stack" control line is enabled.

4) *Scratch Pad*: The 4004 has sixteen 4-bit registers it can use as general purpose memory. This is so that it is able to perform basic functions without the need for any external RAM chips. It is known as the Scratch Pad (Figure 10). Each Scratch Pad location can be addressed individually or as a pair of registers for storing 8-bit values. The Scratch Pad has its own address register. When the Scratch Pad is in pair-mode the highest order 3-bits of the address are used to select a register pair. The instruction then places the value of each register of the pair onto the bus sequentially. When the Scratch Pad is being individually addressed the full address is looked at and the control signal for the correct register is generated. This is done with a separate "Scratch Pad Register Select" control line which is fed into an AND gate with the least significant bit from the Scratch Pad address register. This then generates a "Word Select" line which in turn activates either the first or

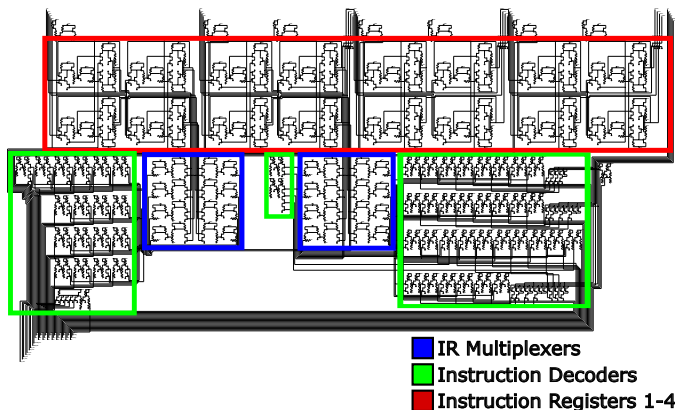


Fig. 5: Circuit diagram for the Instruction Register with colour-coded sub-circuits

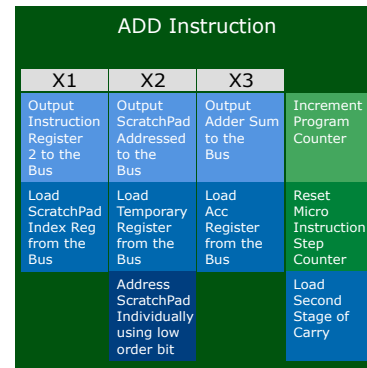


Fig. 6: Diagram showing the control lines that need to be activated per step to execute the ADD instruction

the second word of the specified pair of registers.

5) *Bus*: The internal data bus on a tri-state PMOS 4004 carries three states (logic high, logic low, and high impedance). The voltage-follower output stage of the SiC JFET RTL primitives used here does not implement a clean high-impedance state, so each producer of the four-bit internal bus is instead connected through what is effectively a multi-input OR: if any enabled producer asserts logic high, the bus reads logic high, otherwise it reads logic low. Bus contention is avoided at the micro-instruction-control level rather than at the electrical level – only the intended producer is enabled by the step counter during each micro-instruction phase, and the OR-style combining is used to obtain stable RTL voltage levels rather than the high-impedance behaviour of the original. The connection from the bus to each circuit's input is conventional, since these connections all terminate at JFET gates and so do not exhibit the same voltage-combining problem at the producing side. This deviation preserves the software-visible value placed on the modelled internal data bus in the tested programs (Section IV), but it is not pin-level equivalent to the original 4004 external bus protocol; see the definition of "4004-compatible" at the start of Section II.

6) *Pins and Communication*: The 4004 has a number of pins used to interface with external components (Figure 11). There are four data pins which are used to transfer opcodes and data between the CPU and the external bus ($D_{0,1,2,3}$). There are four CM-RAM pins which are used to address the RAM chips ($CM - RAM_{0,1,2,3}$). When the designate command line (DCL) instruction is executed the value that is in the accumulator is loaded into the CM-RAM register. These pins are "1 Hot" due to the fact that each RAM chip only has one CM-RAM input pin. However, the user is able to put an external de-multiplexer in between the CM-RAM pins and the input of the RAM chip to allow for the full address

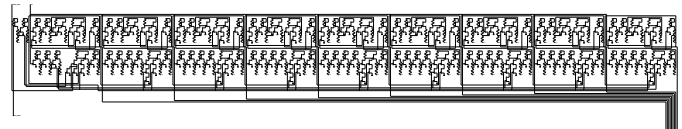


Fig. 7: Circuit diagram for the Micro-Instruction step counter, composed of a series of 1-bit registers

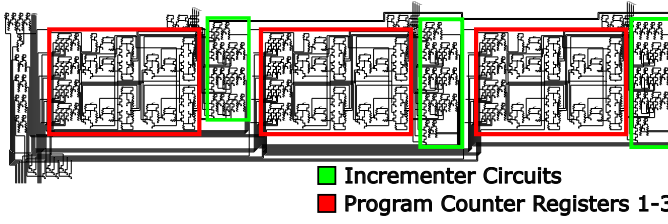


Fig. 8: Circuit diagram for the Program Counter with colour-coded sub-circuits

space to be used rather than just the maximum of four chips as default. The CM-ROM pin is used to communicate with the ROM and the RAM. This pin will go high at specific points in the instruction cycle to indicate various things such as a ROM/RAM instruction. Finally the SYNCH pin which is used to synchronise the operations of all chips in a larger circuit.

III. METHOD

Validation proceeded hierarchically: primitive gates were first checked in LTspice, then composed into sequential sub-circuits and CPU subsystems before full-processor integration. The first sequential subcircuit was the 1-bit synchronous D-type flip-flop. The synchronous part of this simply means that it only latches the D value in on the clock pulse. First the flip-flop was built using discrete logic gates, however, it was noticed that some gates could be reduced using condensed boolean gates as described in the logic primitives section (Section II-A). This was done until the circuit could not be condensed any further due to race conditions and the path for individual signals becoming unbalanced compared to other paths through the circuit. This unbalancing occurred due to the layout of the logic circuit for a D-type flip-flop. The output is taken from a pair of cross-connected gates meaning that if one gate was reached by the signal significantly before the other gate, the output would not latch correctly (Figure 12).

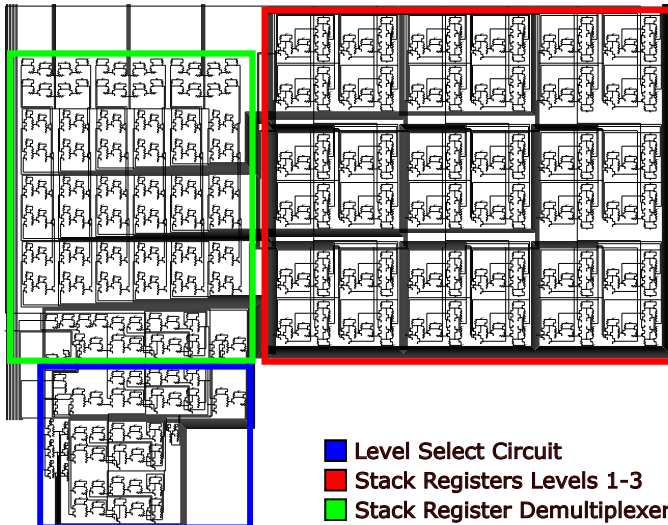


Fig. 9: Circuit diagram for the Stack with colour-coded sub-circuits

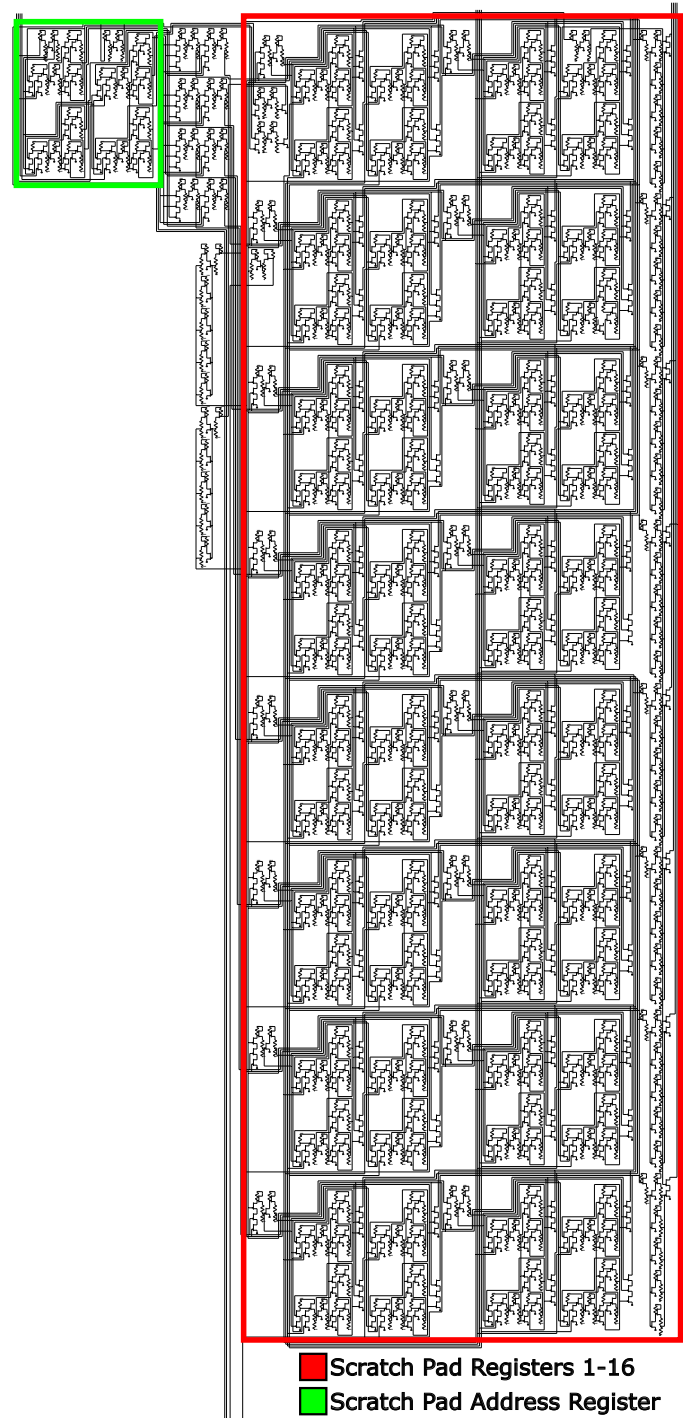


Fig. 10: Circuit diagram for the Scratch Pad with colour-coded sub-circuits

D flip-flops were used for storage in this CPU as 6-transistor (6T) static RAM (SRAM) and 1-transistor (1T) dynamic RAM (DRAM) proved challenging to implement. There are 146 bits of storage in the 4004; with the current 15-JFET D flip-flop, this accounts for 2190 of the processor's transistors. The full per-subsystem JFET budget of the implementation is given in Table III; the scratch pad and stack together consume 1563 and 817 JFETs respectively, dominating the total of 5546. This total is regenerated from the released schematic

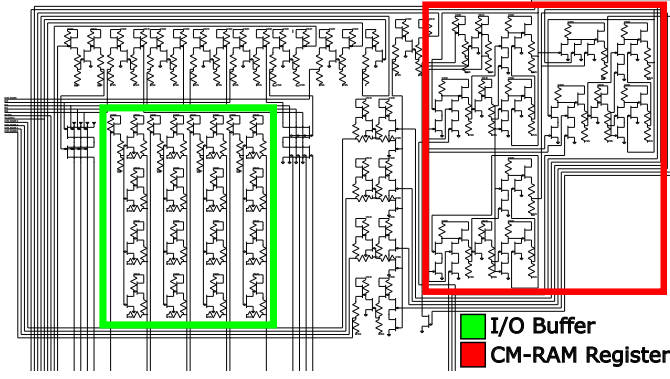


Fig. 11: Circuit diagram for the pins and associated logic with colour-coded sub-circuits

TABLE III: Transistor count of the SiC JFET 4004-compatible processor, per subsystem at full capacity (all sixteen scratch-pad registers and three stack levels included). Numbers are derived directly from the LTspice schematics referenced by `cpus/4004/config.py`; re-running `tools/count_transistors.py` regenerates this table from the current schematic state.

Subsystem	JFETs	% of total	Resistors
Arithmetic logic unit	601	10.8%	540
Instruction register	680	12.3%	642
Micro-instruction decoder	851	15.3%	657
Control logic	28	0.5%	24
External pins and bus	140	2.5%	159
Program counter	503	9.1%	429
Step counter	363	6.5%	351
Scratch pad (16 × 4-bit)	1563	28.2%	1230
Subroutine stack (3-level)	817	14.7%	693
Total	5546	100.0%	4725

metadata using `tools/count_transistors.py`, which walks the `SYMBOL njf` declarations in every component file referenced by the CPU configuration. The count differs slightly from the preliminary hand count of 5328 in earlier reports of this work, which understated the scratch-pad and stack subsystems. Development of 1T DRAM would, by inspection, reduce the total transistor count by roughly 2000 (excluding any RAM refresh circuitry), making storage technology the highest-leverage target for further reduction.

Each sub-system of the CPU was built separately at first in order to decrease simulation times. The system was tested using the following method. Each control line was connected to a voltage source utilising a Piecewise linear (PWL) file. A PWL file is a type of file used by LTspice which specifies the voltage at specific time points. Each control line was listed in a spreadsheet along with register values if appropriate. The necessary voltage level for each control line was then plotted on a timeline to "program" the circuit to perform operations. The PWL files were then generated using Matlab from the list of times the control line needed to be high. The final state of the sub-system could then be compared to the expected state calculated from the spreadsheet. Once each sub-system had been verified as correctly functioning they were assembled into the full CPU by means of a Python script

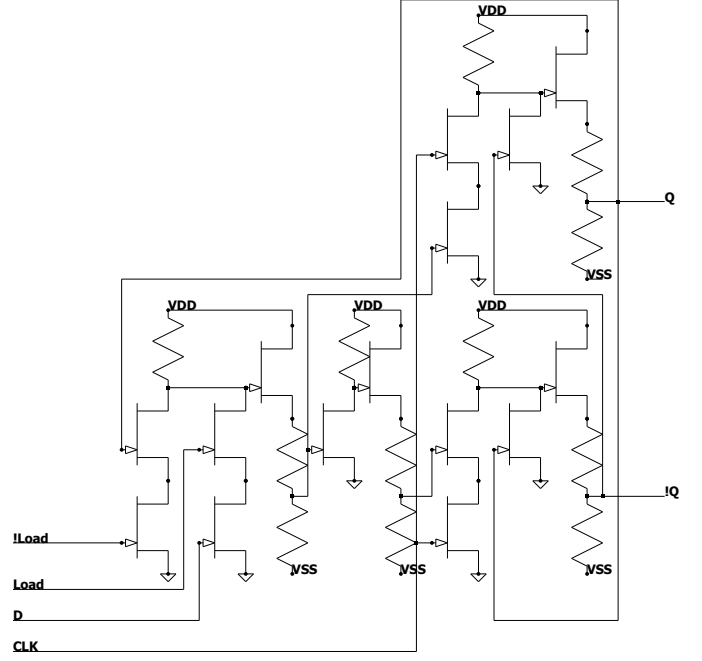


Fig. 12: Circuit diagram for synchronous D-type flip-flop with transistor reductions using combined boolean logic gates

which collated each sub-system and renamed any duplicated component names. This was then tested by connecting the bus to voltage sources and emulating the ROM by feeding it data when it expected to receive data from the ROM. This approach was sufficient for deterministic subsystem tests, but could not validate programs whose control flow depended on CPU-computed state, since every bit returned to the CPU had to be hard-coded beforehand.

A. ROM

In order to fully test the capabilities of this CPU it was decided that a ROM needed to be designed and built (Figure 13). The 4001 functions in a similar way to the 4004. It uses a micro-instruction counter to see where it is in the instruction cycle and also has an instruction register which is used for some ROM specific instructions. The ROM receives an address from the CPU at times A_1 and A_2 . It then receives a chip select address at time A_3 , which determines which of the 4001 chips is being addressed, along with CM-ROM going high. Each 4001 has a specific identity which is hard-wired during production. If the value that is wired into the 4001 matches the address received at A_3 the 4001 then outputs the values stored at the address received from $A_{1,2}$ during M_1 and M_2 otherwise it is deactivated until the next fetch cycle. If a ROM/RAM instruction is detected by the ROM, the instruction register is loaded from the external bus during X_1 . The ROM has two instructions as implemented in this paper, write ROM port (WRR) and read ROM port (RDR) which are used to read and write to the I/O pins on the ROM chip. This is performed during X_2 unless no instruction was received during X_1 in which case the execution cycle is reset ready for the next address to be received. Storage in the ROM was implemented using a voltage source linked to a PWL file for each bit of

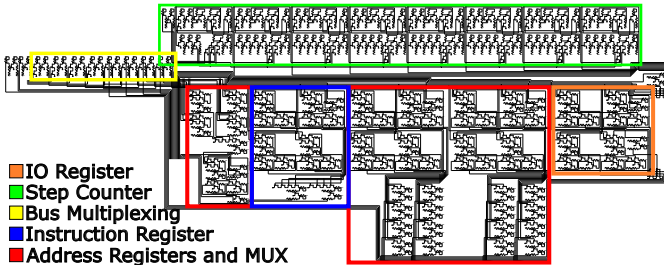


Fig. 13: Circuit diagram for computational part of Intel 4001 with colour-coded sub-systems

every address. When the ROM is addressed the value defined by the PWL file for that address is sent out onto the bus. This was done so that each bit would have a specific PWL file which could be written to, allowing for the ROM to be programmed.

B. Assembler

An assembler converts the mnemonics of the instructions from text to binary and converts the operands from hexadecimal or decimal to binary. Thus `FIM 1 0 11` is converted to `0010 0010 0000 1011`. The assembler is a two-pass implementation. The first pass walks the source text and records, for each label, the ROM byte address at which the next instruction will be emitted, advancing a running address counter by one byte per single-word instruction and by two bytes per two-word instruction (`JCN`, `FIM`, `JUN`, `JMS`, `ISZ`). The second pass emits the byte stream, with each label reference resolved to the recorded 12-bit ROM byte address rather than to a source-text line number. The assembled program is written to a `.bin` file – one 8-bit ROM word per line, in ROM-address order – and to a parallel `.hex` file. The assembler is written in Python.

C. PWL File Generation

The `.bin` file is used by another Python script to generate the PWL files that populate the voltage sources inside the 4001-style ROM model. Each PWL file encodes the fixed contents of a single ROM bit cell – either a constant logical high (-0.8 V) or a constant logical low (-3.6 V) – and is named `0.txt`, `1.txt`, ... in ROM-bit order. Address decoding inside the ROM model selects the addressed byte during the simulated M_1/M_2 phases, so branch and subroutine behaviour during simulation is determined by the CPU state and the ROM address path rather than by an externally scripted bus trace; the PWLs do not encode a precomputed execution trace.

D. Testing and Benchmarking

With a full 4001 and a full 4004 modelled in LTspice, complete programs can be run and their state compared, instruction by instruction, against an independent Python reference emulator (`tools/rom_emulator.py` in the repository). The reference emulator implements the documented 4004 ISA,

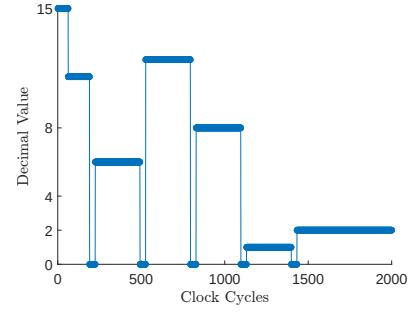


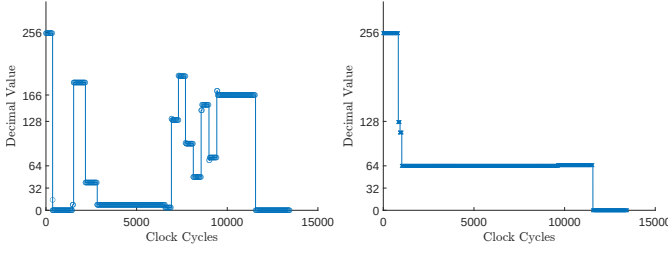
Fig. 14: Figure showing the decimal value in Scratch Pad Register 0 for the operation 11 & 6. This illustrates the first input (11) being shifted left 4 times to compare the highest order bit to the second input (6) with the final result of 2 being loaded in just before 1500 clock cycles

including the carry rules for `ADD/SUB`, the rotate-through-carry semantics of `RAL/RAR`, and the three-level stack handling of `JMS/BBL`. Two test programs are run: the bitwise `AND` subroutine from the MCS-4 programming manual [12] (a 30-byte routine that exercises `XCH`, `RAL/RAR`, `ADD`, `JCN` and `JMS/BBL` — Figure 14) and a 16-bit floating-point multiplication whose execution trace covers 451 dynamically executed instructions and exercises the full ALU through a deeper subroutine call tree. For each program the LTspice CPU is allowed to settle past its power-up transients, the Python emulator is synced to the LTspice register state at a documented sync point, and forward execution from that point compares (`ACC`, `CY`, $R_0, \dots, R_{15}$) at every instruction boundary. The headline pass/fail counts reported in Table IV use `-resync never`: any cascading divergence after the first mismatch is exposed rather than papered over. A separate `-resync on-divergence` diagnostic mode re-aligns the emulator to the LTspice state after each mismatch and is used only to localise independent local errors; it is not used to claim end-to-end program correctness. Section IV presents the resulting per-program diff summaries.

1) *Waveform Analyser and Reference Emulator*: The LTspice `.raw` file is parsed directly by a Python tool (`tools/trace_synced.py`) which thresholds the per-bit register node voltages at -2.5 V to recover digital values, samples them at 95% through each `micro1` phase (after the `IR` and accumulator have settled), and emits one nibble per register per instruction boundary. A 4004 ISA-level reference emulator written in Python (`tools/rom_emulator.py`) is run in lockstep on the same machine code. Implementation quirks of the original 4004 — the incrementer not affecting the accumulator, the LSB convention for `FIN/JIN`, the rotate-through-carry semantics — were validated against an online JavaScript emulator [17] during development and against the contemporary MCS-4 manual [12].

E. Experimental Validation of Simulation Models

Physical validation was performed on `NAND` and `NOR` primitives implemented using the available SiC JFET integrated circuit (IC) and a copper circuit board with through-hole components. These were connected to a power supply, a



(a) Significand which is calculated to be 166 in decimal (b) Exponent which is calculated to be 65 in decimal

Fig. 15: The waveforms of the registers during the computation of the product of 1.4375×3.60937 . Plots 15a and 15b show the values of the significand and exponent respectively during the operation

TABLE IV: Program-level LTspice/emulator co-validation. The compared state vector is $(ACC, CY, R_0, \dots, R_{15})$. The headline pass/fail counts are computed without resynchronisation after the documented initial sync point; a separate resync-on-divergence mode (Section III) is used only for diagnostic localisation and is not used for the values reported here. “Boundaries” is the number of instruction boundaries compared after the sync point; “mismatches” is the number of those boundaries at which any element of the compared state vector differed between the LTspice CPU and the Python reference emulator.

Program	Bytes	Dyn. instr.	Boundaries / Mismatches	Final result
AND subroutine	30	TBD	TBD/TBD	$R_0 = 0010$
FP multiplication	51	TBD	TBD/TBD	5.1875

function generator, and an oscilloscope. They were then tested at a range of frequencies from 1 Hz to 1 MHz. The data from this experiment was then used to characterise the gates. The inputs from the function generator were taken and a Matlab script was written to convert these into a PWL file which could then be used as the inputs for the simulated gates to get an exact comparison between experimental and theoretical.

IV. RESULTS AND DISCUSSION

Table IV collects the headline results of the work. Each row is detailed in the subsections that follow, and the limitations of each claim — in particular the legacy nature of the NAND/NOR measurements and the room-temperature device model — are discussed in Section V.

The primitive-gate physical evidence reported separately in Section IV below covers the NAND and NOR primitives only and supports the gate-level LTspice model rather than the full processor.

A. Program Testing

1) *AND Subroutine*: From the MCS4 programming manual an AND sub-routine (Listing 4) was run on the CPU and its results compared to the emulator. Figure 14 shows the functioning of the AND sub-routine. The input A, 1011 (11), is loaded into Scratch Pad register 0 and subsequently shifted left

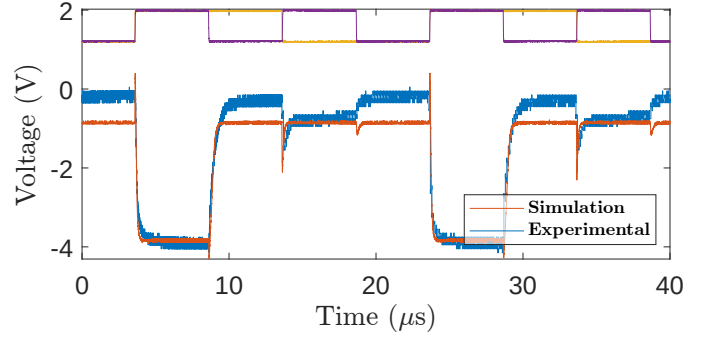


Fig. 16: Experimental and simulation output voltages of a NAND gate at 100 kHz, the operational frequency of the CPU as simulated in this paper. Input voltages are scaled and offset to the top of the figure. Raw oscilloscope traces from the original measurement setup are not retained; the 1.5–2 V baseline shift between measurement and simulation is discussed in Section V.

to extract the highest order bit into the carry. This is done four times, from 1011 (11) to 0110 (6) to 1100 (12) to 1000 (8) and the final output value, 0010 (2), is loaded in after roughly 1400 clock cycles. The value starts at 15 as each register defaults to being high when the CPU is first initialised. From the plots we can see that 6 (0110) AND 11 (1011) gives an output of 2 (0010) which is correct.

2) *Floating Point Multiplication Subroutine*: The floating point system implemented takes the following form.

$$(-1)^{\text{signbit}} \times 1.\text{ssssssss} \times 2^{\text{eeeeeeee}-63} \quad (1)$$

where s denotes a significand bit and e denotes an exponent bit. The processor-level functionality is further demonstrated by the computation 1.4375×3.609375 , which gives 5.1875 after rounding in the implemented floating-point representation.

The implemented floating-point format rounds the exact product 5.18847... to 5.1875, as shown in Table V. Explanations of the algorithm can be found in Appendix A. The final values in the registers in Figures 15a and 15b being zero is

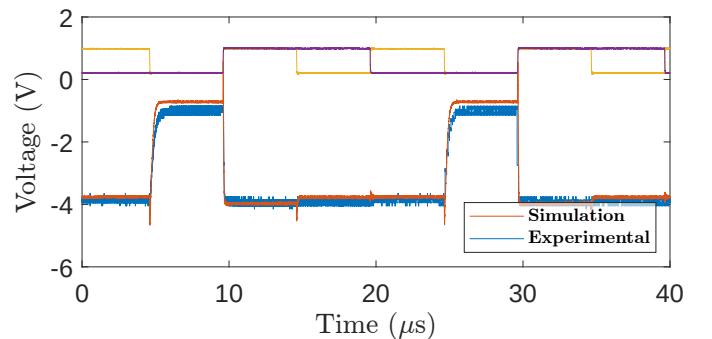


Fig. 17: Experimental and simulation output voltages of a NOR gate at 100 kHz, the operational frequency of the CPU as simulated in this paper. Input voltages are scaled and offset to the top of the figure. Raw oscilloscope traces are not retained; the 1.5–2 V baseline shift is discussed in Section V.

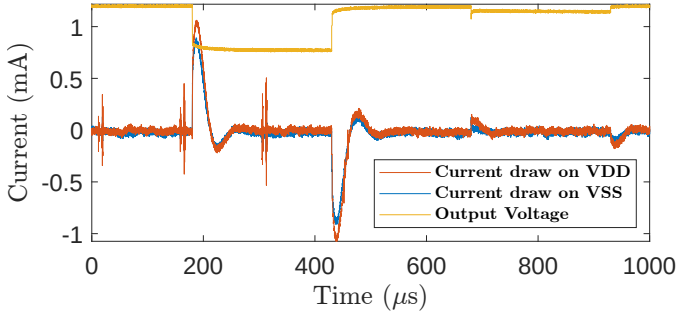


Fig. 18: Plot showing current draw for NAND gate at 2 kHz, the highest of the sampled frequencies that exhibits zero current draw unless the output state is transitioning. The gate output voltage has been scaled and offset to the top of the diagram.

due to an LTspice simulation error which occurred after the program had finished running.

B. Experimental Data

The logic gates were tested at input frequencies from 1 Hz up to 800 kHz (NOR) and 1 MHz (NAND). Figures 16 and 17 show the 100 kHz operating point at which the CPU is simulated, and Figures 21 and 20 show the highest tested frequency for each gate. Above 800 kHz the NOR output swing collapses because the rise time exceeds the half-period, and the gate can no longer respond to its inputs. For the tested two-input primitives, the NOR gate is therefore the measured limiting case, with adequate output swing up to 800 kHz; the NAND gate continues to switch cleanly to 1 MHz. Both bounds are comfortably above the 100 kHz simulated CPU clock.

TABLE V: Floating Point Number Values Used in Subroutine

Binary	Decimal	Exponential Form	Value
10111000 00111111	184 63	1.4375×2^0	1.4375
11100111 01000000	231 64	1.8046875×2^1	3.609375
10100110 01000001	166 65	1.296875×2^2	5.1875

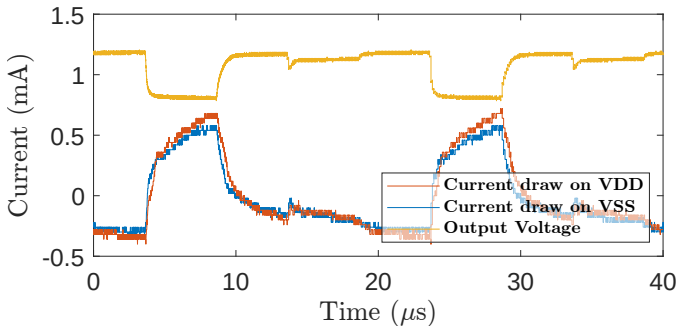


Fig. 19: Plot showing current draw for NAND gate at 100 kHz. This is the operational frequency of the CPU as simulated in this paper and shows that the device will always draw current regardless of input states at these frequencies. The gate output voltage has been scaled and offset to the top of the diagram.

TABLE VI: Register values for Figures 15a and 15b

Clock Cycle	Significand	Exponent	Decimal
0	255	255	1.9921875×-2^{64}
370	0	255	0
828	0	127	0
924	0	112	0
1020	0	64	0
1527	184	64	2.875
2178	40	64	0.625
2830	8	64	0.125
6580	4	64	0.0625
6968	130	64	2.03125
7356	193	64	3.015625
7744	96	64	1.5
8128	48	64	0.75
8608	152	64	2.375
9039	76	64	1.1875
9472	166	64	2.59375
9640	166	65	5.1875

The primitive-gate measurements show that the measured SiC JFET RTL gates switch comfortably above the 100 kHz CPU simulation clock; this should not be read as a measured full-processor $f_{\max}$, since the complete CPU was not fabricated and the critical paths are only evaluated in LTspice. For reference, the original Intel 4004 silicon was specified at ~ 740 kHz peak, with the commonly quoted 400 kHz referring to the typical operating frequency rather than the maximum.

The oscilloscope probes introduced scaling by a factor of two, necessitating post-experimental adjustments. There was a 1.5 - 2 V baseline shift in the data compared to what was expected from the simulations, the origins of which are currently under investigation. Some of the difference was due to resistor inaccuracies, however, this cannot account for the entire 2 V shift. Further, it can be seen that as the frequency increases this offset increases, starting at approximately 1.5 V at 2 Hz going up to approximately 2 V at 100 kHz. This indicates that it is some function of the JFETs themselves. This change in output voltage levels as a function of frequency was only observed in the NAND gate. The NOR gate retained its 1.5 V shift for the entire range of examined frequencies. This could be due to differences in the JFETs used between the NAND and NOR gates rather than due to gate topology.

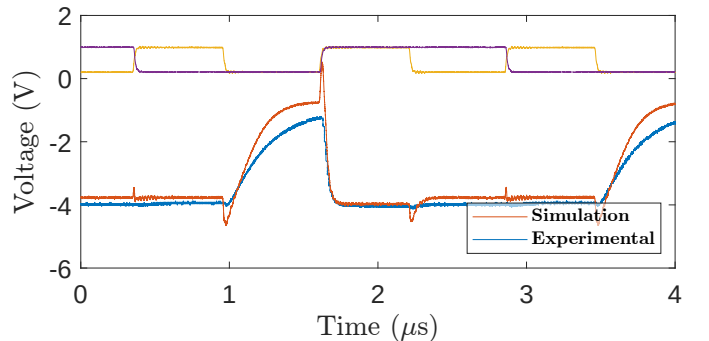


Fig. 20: Experimental and simulation output voltages of a NOR gate at 800 kHz, the maximum frequency at which the NOR gate produced adequate output peaks. Inputs scaled and offset as before. Raw oscilloscope traces are not retained; the 1.5–2 V baseline shift is discussed in Section V.

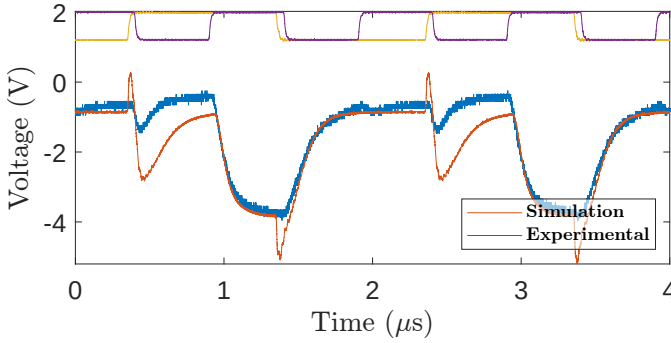


Fig. 21: Experimental and simulation output voltages of a NAND gate at 1 MHz, the maximum frequency at which the NAND gate was tested. Inputs scaled and offset as before. Raw oscilloscope traces are not retained; the baseline shift, which grows with frequency for the NAND gate, is discussed in Section V.

Further investigation into the nature of these discrepancies is necessary to determine their source, be it manufacturing variability or the LTspice model itself. The voltage levels and resistor values used should be explored in greater depth to determine optimum performance for specific applications.

The current draw demonstrated by the NAND gate at lower frequencies displays CMOS like qualities where it is only actively drawing current on the transition between output states as shown in Figure 18. This behaviour was not apparent at higher frequencies as shown in Figure 19. This is an unexpected result and should be investigated further. The current draw on each voltage source (positive supply voltage: 24 V, negative supply voltage: -20 V) was about 0.5 mA per logic gate which consisted of 3 transistors. Extrapolating from this, the peak current draw for the full CPU of 5546 transistors would be of order 900 mA. Due to the CMOS like nature of the current draw at lower frequencies, if entire systems of the CPU such as the lower levels of the stack or some scratchpad registers are not being used, the current draw would be lower.

V. LIMITATIONS

Three classes of limitation deserve explicit acknowledgement.

Measured primitive gates exhibit a 1.5–2 V baseline offset relative to the simulated waveforms (Figures 16, 17, 20, 21). The offset is approximately constant for the NOR gate across the entire measured frequency range, and grows mildly with frequency for the NAND. Resistor tolerance is too small to account for the magnitude; this points instead either to manufacturing variability of the SiC JFET die used (a single integrated circuit was the source of devices for both gates) or to a residual modelling gap in the JFET parameter card used for the LTspice simulation. The waveform *shape* (transitions, settling, and frequency response) matches the simulation closely over 1 Hz to 800 kHz for the NOR and 1 Hz to 1 MHz for the NAND, which is the basis on which the simulated CPU is asserted to be functionally representative; the absolute offset is an open discrepancy and should not be read as a confirmation of the simulation’s quantitative voltage levels.

The full processor is not fabricated. All system-level claims rest on LTspice transient simulation of the complete 5546-transistor design, cross-validated against an independent Python reference emulator at every instruction boundary (Section III). The strongest experimental evidence in this paper is for the primitive gates only. A fabricated CPU is the obvious next step, but the dominant transistor budget — D flip-flop storage rather than combinational logic — would benefit substantially from a single-transistor DRAM cell before fabrication is attempted.

The LTspice device model used is at room temperature. The CPU simulations use a SiC JFET parameterisation supplied for this project, evaluated at 27 °C; high-temperature behaviour is by extrapolation only. The parameter card itself appeared in the publicly released MEng report from which this work descends and is shipped with the reproducibility package on that basis; the underlying device model belongs to its original owner, and any redistribution beyond academic reproduction should obtain permission accordingly. The radiation-hardness and high-temperature motivation for SiC JFET computing remains; what this work demonstrates is that an Intel-4004-compatible architecture is feasible in a SiC JFET RTL logic family, not that the specific design operates at the temperatures motivating it. A temperature-scaled re-simulation of the worst-case combinational chain and the noise-margin corner sweep is a natural follow-up, deferring the chip-level fabrication question while strengthening the temperature claim.

A full fanout / noise-margin / critical-path characterisation of the integrated design remains future work. The 100 kHz CPU claim rests on LTspice transient execution of the reported test programs rather than on a systematic worst-case critical-path extraction across all 46 instructions, and the primitive-gate frequency results in Section IV should not be read as a substitute. The corner-sweep test benches that would produce noise-margin and timing tables – under $\pm 20\%$ resistor variation and $\pm 2\text{V}$ supply variation – are included in the reproducibility package (analysis/sweeps/) but not yet executed; the corresponding result tables will be added in revision.

VI. CONCLUSION

This paper has presented a transistor-level SiC JFET implementation of an Intel-4004-compatible 4-bit microprocessor, comprising 5546 JFETs (Table III) across the ALU, instruction register and decoder, program counter, three-level stack, and sixteen-register scratch pad; a compatible 4001-style ROM model is used for program execution and is not included in this JFET count. The simulated CPU design is supported by two independent forms of evidence: register-level program co-validation against a Python reference emulator at every instruction boundary, demonstrated on a 30-byte bitwise AND subroutine and a 16-bit floating-point multiplication producing a 451-instruction execution trace; and measured waveform-shape and switching-frequency consistency of NAND and NOR primitives built from a discrete SiC JFET integrated circuit, characterised from 1 Hz to 800 kHz (NOR) and 1 MHz (NAND).

The largest practical constraint on further work is the storage cost of D flip-flops: 146 bits of scratch-pad and stack storage account for 2190 JFETs of the 5546 total. The design therefore remains substantially larger than the original NMOS 4004, primarily because the present storage implementation uses 15-JFET D flip-flops rather than 1T DRAM cells. A SiC JFET DRAM cell would close most of this gap. The remaining open discrepancy in this paper — the 1.5–2 V baseline offset between measured and simulated primitive gates — is a candidate next investigation, and is most cheaply pursued by repeating the NAND/NOR measurement with multiple device samples to separate manufacturing variability from a residual modelling gap. A temperature-scaled re-simulation of the worst-case combinational chain and the noise-margin corner sweep is a complementary follow-up that strengthens the high-temperature claim without requiring fabrication. The fabrication of the full 4004-compatible processor in SiC JFET is the eventual target; the work presented here establishes that a complete architecture-scale design and a validation methodology now exist to support that step.

ACKNOWLEDGEMENTS

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APPENDIX A

```

1  SIGNBITS CLC
2  LD 6
3  RAL
4  TCC
5  XCH 2
6  LD 6
7  RAL
8  CLC
9  RAR
10 XCH 6
11 CLC
12 LD 10
13 RAL
14 TCC
15 XCH 3
16 LD 10
17 RAL
18 CLC
19 RAR
20 XCH 10
21 BBL 0

```

Listing 1: Assembly Code for SIGNBITS function

```

1  START LDM 15
2  XCH 4
3  LDM 4
4  XCH 5
5  LDM 4
6  XCH 6
7  LDM 7
8  XCH 7
9  LDM 10
10 XCH 8
11 LDM 0
12 XCH 9
13 LDM 4
14 XCH 10
15 LDM 11
16 XCH 11
17 JMS MULT
18 NOP

```

Listing 2: Assembly Code for initialising registers for multiplication function

The way the multiplication algorithm (Listing 3) works is two separate computations. First the sign bit is extracted from the exponent using the SIGNBITS Subroutine (Listing 1). Then the two exponents are added together, however this means that the offset of 63 has been added twice and needs to be subtracted to get the final value. Next the significands need to be multiplied. This is achieved by checking the lowest order bit of the multiplier. If it is high, the multiplicand is added to the output register, the multiplicand is shifted to the left and the multiplier is shifted to the right. This then repeats until the entire multiplier has been checked and the final answer can be read. There are some tricks that have been done to preserve accuracy of the final answer such as using 4 registers to store the multiplicand and the product allowing for 16 bits of accuracy. The final step shifts the entire product until it detects that there are no 1s in the two registers used for the highest order bits. If the product is shifted more than 8 times it indicates that the product required normalising, shifting so that there is a 1 in the highest order bit. If this occurs, 1 needs to be added to the exponent for each time the product is shifted past 8 times.


```

1  MULT FIM 6 0 0
2  JMS SIGNBITS
3  LD 7
4  ADD 11
5  XCH 15
6  LD 6
7  ADD 10
8  XCH 14
9  CLC
10 LDM 15
11 XCH 0
12 LD 15
13 SUB 0
14 XCH 15
15 CMC
16 LDM 3
17 XCH 0
18 LD 14
19 SUB 0
20 XCH 14
21 CLC
22 LD 2
23 ADD 3
24 RAL
25 RAL
26 RAL
27 CLC
28 ADD 14
29 XCH 14
30 FIM 0 0 0
31 FIM 1 0 0
32 BCHECK LD 8
33 JCN 12 IF1
34 LD 9
35 JCN 12 IF1
36 JUN RETURN
37 IF1 CLC
38 LD 8
39 RAR
40 XCH 8
41 LD 9
42 RAR
43 XCH 9
44 JCN 10 IF2
45 CLC
46 LD 13
47 ADD 5
48 XCH 13
49 LD 12
50 ADD 4
51 XCH 12
52 LD 3
53 ADD 1
54 XCH 3
55 LD 2
56 ADD 0
57 XCH 2
58 IF2 CLC
59 LD 5
60 RAL
61 XCH 5
62 LD 4
63 RAL
64 XCH 4
65 RAL
66 LD 1
67 RAL
68 XCH 1
69 LD 0
70 RAL
71 XCH 0
72 JUN BCHECK
73 RETURN LD 2
74 JCN 12 IF3
75 LD 3
76 JCN 12 IF3
77 JUN MULTEND
78 IF3 LD 2
79 RAR
80 XCH 2
81 LD 3
82 RAR
83 XCH 3
84 LD 12
85 RAR
86 XCH 12
87 LD 13
88 RAR
89 XCH 13
90 CLC
91 JUN RETURN
92 MULTEND BBL 0

```

Listing 3: Assembly Code for floating point multiplication subroutine

APPENDIX B

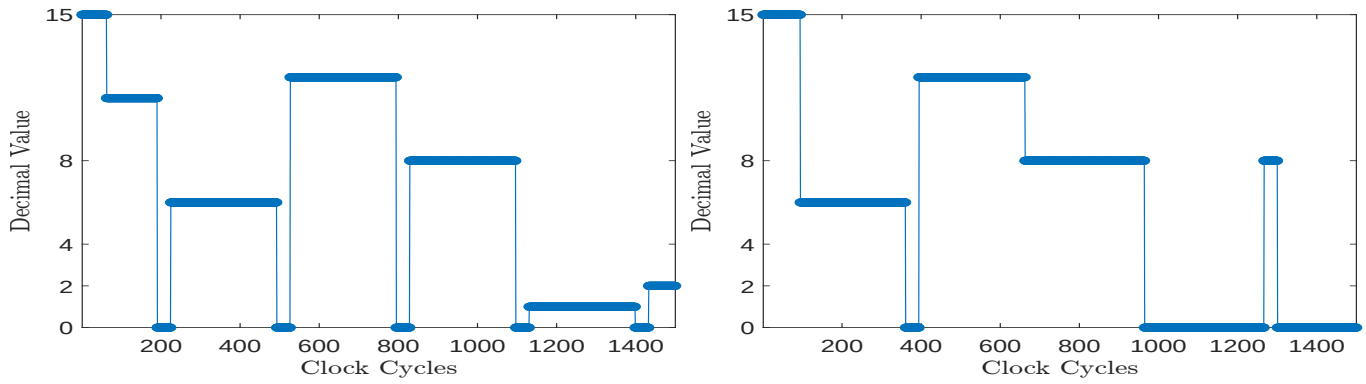
```

1  JUN START
2  AND FIM 1 0 11
3  Loop LDM 0
4  XCH 0
5  RAL
6  XCH 0
7  INC 3
8  XCH 3
9  JCN 4 Done
10 XCH 3
11 RAR
12 XCH 2
13 XCH 1
14 RAL
15 XCH 1
16 RAR
17 ADD 2
18 JUN Loop
19 Done BBL 0
20 START LDM 11
21 XCH 0
22 LDM 6
23 XCH 1
24 JMS AND
25 NOP

```

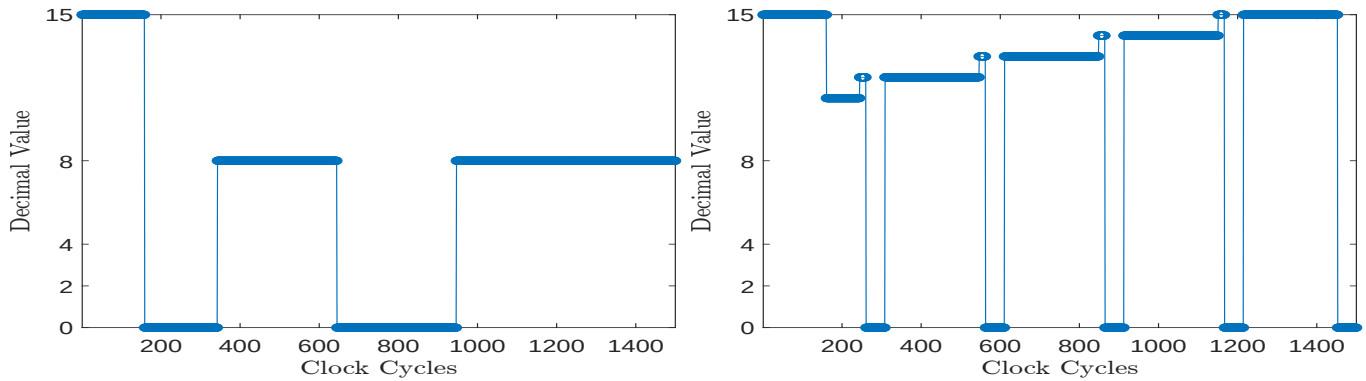
Listing 4: Assembly Code for AND Subroutine

The AND subroutine produces the AND of two input words by placing the bits in the leftmost position of the accumulator 24a and register 2 23a, respectively, and zeroing the rightmost three bits of the accumulator and register 2. Register 2 is then added to the accumulator, and the resulting carry is equal to the AND of the two bits. Scratch Pad register 3 (Figure 23b) is used as the counter, counting up from 11. When this overflows from 15 to 0 the program breaks and returns.



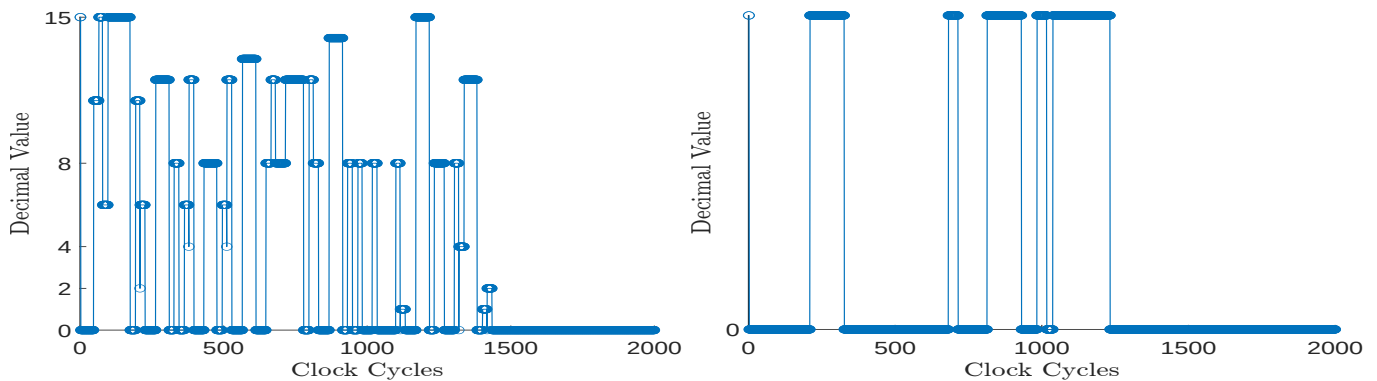
(a) Scratchpad 0 value, containing the first input word which is shifted (b) Scratchpad 1 value, containing the second input word which is left four times. This is the register the function returns the output shifted left three times. The shifts are as follows, 0110 to 1100 to 1000. 1000 to 0000. The returned value is then placed in and shifted once, 0001 to 0010.

Fig. 22: Scratchpad Pair 0, the input and output registers.



(a) Scratchpad 2 value, this is working memory for the program. The (b) Scratchpad 3 value, this is the counter for counting the number of highest order bit from one input is put into here and added to the times register 0 has been shifted. This starts at 11 and the program accumulator which contains the high order bit from the other input. breaks when the value overflows from 15 to 0.

Fig. 23: Scratchpad Pair 1



(a) Accumulator register value

(b) Carry Flag register value, when the highest bit of word one and word two are added together the carry is equal to the AND of the two bits

Fig. 24: Accumulator and Carry Flag register values